A Theoretical Analysis

We now show that Splice terminates in a finite number of profit evaluations and returns a solution with a formally characterized optimality property.

A.1 Splice Convergence

We begin with defining κ-local optimum and use it to establish convergence conditions.

Definition A.1 (κ-local optimum). A subset $S \subseteq \mathcal{S}$ is a *κ-local optimum* of $\mathcal{P}_{\mathcal{T}}$ if no swap of up to κ sources improves profit, i.e. for all $R \subseteq S$ and $D \subseteq \mathcal{S} \setminus S$ with $|R| = |D| \le \kappa$:

$$\mathcal{P}_{\mathcal{T}}\big((S \setminus R) \cup D\big) \le \mathcal{P}_{\mathcal{T}}(S). \tag{2}$$

Theorem A.2 (Convergence). *Let $\mathcal{P}_{\min}$ and $\mathcal{P}_{\max}$ be finite lower and upper bounds on $\mathcal{P}_{\mathcal{T}}$ over $2^{\mathcal{S}}$. Introduce a minimum profit improvement $\varepsilon > 0$ required to accept a swap. For fixed $k_{\max}$ and B, Splice terminates after at most*

$$O\left(B \cdot k_{\max} \cdot n \cdot \frac{\mathcal{P}_{\max} - \mathcal{P}_{\min}}{\varepsilon}\right)$$

profit evaluations. Furthermore, if Splicing exhaustively evaluates all k-swaps (rather than the marginal-gain heuristic), the returned subset is a $k_{\max}$-local optimum of $\mathcal{P}_{\mathcal{T}}$ (Definition A.1).

In our implementation, every strict improvement is accepted; ε serves as an analytical parameter abstracting the minimum representable positive profit increment on a finite dataset.

Proof. Splicing accepts a candidate S' only when $\mathcal{P}_{\mathcal{T}}(S') \ge \mathcal{P}_{\mathcal{T}}(S_{\text{curr}}) + \varepsilon$, so profit increases by at least ε at each accepted step. Since $\mathcal{P}_{\mathcal{T}}$ is bounded in $[\mathcal{P}_{\min}, \mathcal{P}_{\max}]$, the number of accepted steps per FixedSupport run is at most $(\mathcal{P}_{\max} - \mathcal{P}_{\min})/\varepsilon$. At each step, Splicing iterates over $k = 1, \dots, k_{\max}$. For each k, it computes marginal gains $\Delta^-(s)$ for every $s \in S$ ($|S| \le n$ evaluations) and $\Delta^+(u)$ for every $u \in \mathcal{S} \setminus S$ ($n - |S|$ evaluations), then evaluates the single proposed swap (1 evaluation): $O(n)$ evaluations per k, and $O(k_{\max} \cdot n)$ per Splicing call. Over all accepted steps and B initializations, total evaluations are bounded by $O(B \cdot k_{\max} \cdot n \cdot (\mathcal{P}_{\max} - \mathcal{P}_{\min})/\varepsilon)$. Splice terminates when no proposed k-swap improves profit by ε for any $k \le k_{\max}$, i.e. $S_{\text{curr}} = S_{\text{prev}}$ in FixedSupport. If Splicing exhaustively evaluates all $\binom{|S|}{k}\binom{n-|S|}{k}$ candidate k-swaps rather than using the marginal-gain heuristic, this termination condition is equivalent to Definition A.1, and the returned set is a $k_{\max}$-local optimum. The practical algorithm uses marginal gains to guide swap selection, trading the formal local-optimum guarantee for the $O(k_{\max} \cdot n)$ per-step cost. □

Practical implication. We observe in experiments (Section 6) that Splice converges in $319 - 6{,}385$ evaluations (Figure 3), far below the exhaustive worst case of $O(n^{2k_{\max}})$, because the profit landscape has few local optima at the operating $k_{\max} = 5$ and generating the active set with the top-i sources ranked by individual profit already starts the search in high-profit regions of the subset space.

A.2 Submodularity Ratio of Profit Landscape

Theorem A.2 guarantees termination at a local optimum but says nothing about its distance from the global optimum. We characterize the structure of $\mathcal{P}_{\mathcal{T}}$ via the *submodularity ratio* of Das and Kempe [9], used here as a diagnostic.

Definition A.3 (Submodularity ratio [9]). The submodularity ratio of $\mathcal{P}_{\mathcal{T}}$ with respect to ground set U (the universe of items over which $\mathcal{P}_{\mathcal{T}}$ is defined) and parameter κ is

$$\gamma_{U,\kappa} = \min_{\substack{L \subseteq U \\ S \subseteq U \setminus L \\ |S| \le \kappa}} \frac{\sum_{i \in S} \big[\mathcal{P}_{\mathcal{T}}(L \cup \{i\}) - \mathcal{P}_{\mathcal{T}}(L)\big]}{\mathcal{P}_{\mathcal{T}}(L \cup S) - \mathcal{P}_{\mathcal{T}}(L)}.$$

In our setting, $U = \mathcal{S}$, the full set of candidate sources. Das and Kempe's original definition assumes monotone $\mathcal{P}$, under which $\gamma \in [0, 1]$ ($\gamma = 1$ for submodular functions), and show that a greedy solution achieves a $(1 - e^{-\gamma})$ approximation for monotone objectives. Our profit function is non monotone because adding a source is not guaranteed to increase utility; therefore, the formal Das-Kempe interpretation does not apply and the approximation guarantee does not transfer to our setting. We, therefore, report the ratio as a qualitative landscape diagnostic rather than a formal bound, with values > 1 indicating that sum of the individual gains exceeds the combined gain, and values < 0 indicating that the individual and combined gains disagree. We report the results from our empirical estimation in Section A.3.

A.3 Empirical estimation of submodularity ratio

As discussed in Section A.2, for each benchmark, we estimate $\gamma_{\mathcal{S},5}$ – i.e., the submodularity ratio with ground set equal to the full candidate pool $\mathcal{S}$ and parameter $\kappa = 5$ – by sampling 500 random subset pairs (L, S) with $|S| = 5$. For each pair, we compute the ratio of the sum of individual marginal gains to the combined gain. Because the joint gain can be zero or negative in our non-monotone setting (in which case the ratio is undefined or numerically unstable), we filter such pairs and report the median and the fraction of remaining pairs falling into the regimes stated in Section A.2.

Table 9 reports the results. All benchmarks have a median ratio above 1, indicating that substitution dominates the typical landscape: in 53–78% of sampled pairs, the sum of individual contributions exceeds the joint contribution, reflecting overlapping signal across sources. Two patterns vary meaningfully across benchmarks. First, the median itself varies by a factor of 2.3×, from a barely substitution flavored ACSTravelTime (median 1.07) to the strongly substitution flavored ACSPublicCoverage (2.37) and IPO (2.43). Second, the opposite sign fraction ranges from 8% on Amazon to 33% on ACSTravelTime, characterizing how often individual sources have effects of opposite sign to the joint effect. The fraction of sampled pairs with non-positive joint gain - those filtered before ratio computation - ranges from 12% on Wilds (where source diversity makes most multi source additions broadly useful) to 47% on ACSPubCoverage (where source cancellation is most prevalent).

These diagnostic patterns track Splice's empirical advantage over greedy forward selection. On benchmarks with high median and low opposite sign fraction (ACSPubCoverage, IPO, Amazon), the landscape rewards local greedy choices, and greedy reaches profit within 0.5–2.2 points of Splice. On ACSTravelTime, the lowest median benchmark with the highest opposite sign fraction, greedy gets stuck in local optima that 1-step additions cannot escape — Splice exceeds greedy by 11 MSE points. On Wilds, the

Table 10: Paired difference in profit, SPLICE minus the competing method, on identical target splits/seeds. Positive favours SPLICE. p from a paired t-test on per-replicate differences.

Benchmark	vs.	Δ (95% CI)	p
ACSIncome	ALLSRC	$+1.749$ $[1.63, 1.87]$	$< 10^{-4}$
	DSDM	$+0.333$ $[0.15, 0.51]$	0.007
	GRASP	$+0.008$ $[-0.07, 0.09]$	0.836
	SPLICE-DP	-0.063 $[-0.16, 0.04]$	0.709
ACSPubCov	ALLSRC	$+13.690$ $[13.65, 13.73]$	$< 10^{-4}$
	GRASP	$+0.044$ $[0.02, 0.07]$	0.005
	SPLICE-GRAD	-0.060 $[-0.08, -0.04]$	$< 10^{-4}$
ACSTravelTime	ALLSRC	$+14.836$ $[14.48, 15.19]$	$< 10^{-4}$
	GRASP	-0.993 $[-1.26, -0.73]$	$< 10^{-4}$
Amazon	ALLSRC	$+0.0299$ $[0.022, 0.037]$	$< 10^{-3}$
	DSDM	$+0.0063$ $[0.004, 0.009]$	0.001
	GRASP	-0.0001 $[-0.0004, 0.0002]$	0.467
Wilds	ALLSRC	$+0.0215$ $[0.014, 0.029]$	0.001
	DSDM	$+0.0197$ $[0.009, 0.031]$	0.020
	SPLICE-DP	$+0.0110$ $[-0.000, 0.023]$	0.100

Table 9: Submodularity ratio diagnostics across benchmarks. *Substitutes* ($\gamma > 1$, sum of individual gains exceeds joint gain), *Partial-complements* ($0 < \gamma \leq 1$, joint gain exceeds or matches the sum), *Opposite-sign* ($\gamma < 0$, individual gains and joint gain have opposite signs, only possible in non-monotone settings). *% Filtered* reports the fraction of all 500 sampled pairs filtered before γ computation because the joint gain was non-positive.

Benchmark	Median γ	% Sub	% Part-comp	% Opp-sign	% Filtered
ACSIncome	1.97	77	11	11	26
ACSPubCov	2.37	76	11	11	47
ACSTravelTime	1.07	53	14	33	41
Amazon	1.71	78	14	8	24
IPO	2.43	69	10	21	39
Wilds	1.44	69	14	17	12

substitution structure is moderate but the per-source value variance is large (single-source macro-F1 spans 0.05–0.20), so multistart initialization and k-swap together are necessary to navigate the landscape — SPLICE outperforms greedy by 8.66 macro-F1 points, the largest gap in our suite. The submodularity ratio diagnostic is consistent with where SPLICE's mechanism is most beneficial, even though it does not transfer formal approximation guarantees to the non-monotone setting.

B Reference Models for SPLICE-GRAD

SPLICE-GRAD uses a task-specific reference model trained on the union of all source data before gradient extraction. Architectures are selected to be lightweight (so gradient computation is fast) while remaining expressive enough that gradient directions are informative.

Tabular classification (ACSIncome, ACSPubCoverage, SANTOS). An ℓ_2-regularized logistic regression model (`nn.Linear`$(p, 1)$ with sigmoid output), trained with binary cross-entropy for 5 epochs. Gradient dimension: $d_g = p + 1$, where p is the number of input features.

Tabular regression (ACSTravelTime). A single linear layer (`nn.Linear`$(p, 1)$), trained with mean squared error for 5 epochs. Gradient dimension: $d_g = p + 1$.

Image classification (Wilds). A linear head `nn.Linear`(d_{feat}, c) over frozen ResNet-18 backbone features. The head is warm-started with cross-entropy for 5 epochs on the union of all source data; gradients are then recomputed against a differentiable soft-macro-F1 loss [12] so that the direction matches the macro-F1 evaluation metric. Gradient dimension: $d_g = d_{\text{feat}} \cdot c + c$ (full weight and bias gradient).

Text classification (Amazon). A two-layer MLP head `LayerNorm` $\to$ `Linear`(d_{feat}, h) $\to$ `ReLU` $\to$ `Dropout` $\to$ `Linear`$(h, 1)$. The head is warm-started in two phases: first with binary cross-entropy on the union of all source data, then fine-tuned with differentiable soft-F1 loss [12]. We store only the final-layer weight gradient ($d_g = h$) to keep per-query cost low. Target gradients are computed on stratified positive/negative samples and averaged across 5 random seeds to reduce variance [47].

C Paired differences vs SPLICE

Table 10 reports the closest pairwise comparisons against SPLICE on identical replicates. We use the paired difference rather than the marginal intervals of Table 3 because replicate-to-replicate variance is large and shared across methods: on ACSIncome, for instance, every method's profit varies by about 2.5 points across the 30 target splits (sd ≈ 0.6), while the standard deviation of the SPLICE $-$GRASP *difference* is 0.22. Exact ties are common on the small ACS pools, where two searches frequently converge on the same subset.

Wilds GRASP. Table 3 reports GRASP on Wilds without a confidence interval. A single run requires 14,606 model trainings and 12.4 hours; ten seeds would exceed five days of compute on the evaluation machine, so we report the single-seed value and do not claim a significance test against it. The paired comparisons in Table 10 accordingly omit GRASP on Wilds.

Table 11: ACSPubCov source profiles, ordered by solo profit.

State	Sel.	Rows	Mismatch	Shift	Solo	Marginal
CA		729,432	0.211	4.21	62.23	-5.89
ME	✓	23,699	0.254	3.26	61.55	$+0.09$
ID	✓	30,490	0.362	3.65	59.79	$+0.53$
MI	✓	186,896	0.230	3.01	58.05	$+1.81$
MD		93,984	0.279	3.10	57.86	-1.04
GA		194,141	0.358	3.74	57.75	-5.39
FL		367,768	0.337	3.81	57.44	-2.14
MN		86,374	0.260	3.00	55.63	-0.14
NY	✓	356,554	0.183	2.38	54.18	$+7.77$
WI		98,558	0.303	1.54	53.17	-7.40
OH		217,837	0.239	2.30	49.70	-2.45
NE		32,572	0.383	2.80	48.12	-0.06
NC		191,528	0.320	2.32	45.20	-7.88
TX		502,018	0.379	1.69	44.68	-17.97
UT		56,819	0.430	1.84	44.52	-8.99
Mean, selected		149,410	0.257	3.08	58.39	$+2.55$
Mean, discarded		233,730	0.318	2.76	52.39	-5.40

Table 12: Wilds source profiles. Cls: classes present in the source; Cov: fraction of target classes covered.

Src	Imgs	Cls	Cov	Shift	Solo	Marginal
Selected by Splice						
28	3,719	17	0.452	10.06	0.0669	+0.029
2	4,100	24	0.548	8.14	0.0574	+0.062
10	3,690	22	0.516	8.47	0.0549	+0.070
9	2,620	23	0.516	7.92	0.0477	+0.090
20	2,420	15	0.419	9.31	0.0279	+0.038
27	2,400	19	0.065	10.88	0.0222	+0.025
11	3,786	4	0.097	11.42	0.0184	+0.060
15	3,448	4	0.097	10.73	0.0184	+0.052
12	2,394	6	0.129	9.85	0.0184	+0.048
14	7,639	6	0.129	8.96	0.0184	+0.027
1	2,112	3	0.065	12.31	0.0184	+0.016
16	2,573	2	0.065	14.54	0.0184	+0.008
29	2,810	20	0.452	7.63	0.0142	+0.022
24	8,844	15	0.065	9.70	0.0077	+0.016
Discarded						
4	2,802	21	0.516	6.44	0.1095	−0.051
3	4,190	25	0.581	5.98	0.0765	−0.067
13	2,219	18	0.452	6.72	0.0648	−0.033
26	2,260	22	0.516	6.31	0.0632	−0.028
7	2,500	19	0.548	6.15	0.0567	−0.064
0	3,640	21	0.484	6.28	0.0545	−0.068
17	3,982	20	0.548	6.03	0.0515	−0.022
19	3,327	18	0.452	6.87	0.0490	−0.014
22	3,038	14	0.355	7.44	0.0431	−0.009
23	2,220	13	0.355	7.20	0.0334	−0.034
18	4,528	13	0.323	7.35	0.0280	+0.002
25	3,770	25	0.581	5.52	0.0251	−0.022
6	3,290	14	0.419	7.02	0.0207	−0.069
5	3,069	7	0.097	8.11	0.0185	−0.080
21	6,855	2	0.065	8.60	0.0184	−0.014
8	2,840	21	0.484	6.98	0.0162	−0.039
Mean, selected			0.258	9.28	0.0293	+0.040
Mean, discarded			0.423	6.81	0.0456	−0.038

D Per-source profiles

Section 6.2 argues that per-source scores do not predict selection. This appendix gives the evidence one benchmark at a time, each chosen to isolate one of the three failure modes of Section 1: source cancellation on ACSPubCov, feature coverage gaps on Wilds, and label mismatch on Amazon. Throughout, *Shift* is the ℓ_2 distance between source and target feature centroids, *Solo* is profit when training on that source alone, and *Marginal* is the source's contribution to the subset Splice returns. Each table closes with the mean over selected and over discarded sources, which is the comparison the discussion turns on.

D.1 ACSPublicCoverage: source cancellation

Splice keeps 4 of 15 states. The ranking by solo profit and the ranking by marginal contribution disagree at the top: California has the highest solo profit of any state (62.23) and the second-most negative marginal contribution (−5.89), while New York ranks ninth by solo profit (54.18) and contributes the most of any source (+7.77). Selected states average 58.39 solo against 52.39 for discarded ones, a gap far smaller than the gap in marginal contribution (+2.55 against −5.40). Texas shows the mechanism at its most extreme: the second-largest pool of rows in the benchmark, and a marginal contribution of −17.97. Pooling all fifteen scores 51.12 against Splice's 64.81 precisely because contributions of this magnitude cancel.

D.2 Wilds: feature coverage gaps

Wilds inverts the pattern one would expect. Splice keeps 14 of 30 locations, and the selected ones are *worse* on every per-source statistic: lower solo profit (0.029 against 0.046), narrower class coverage (0.258 against 0.423), and larger distance from the target (9.28 against 6.81). Six selected locations share an identical solo macro-F1 of 0.0184 and contain only 2–6 classes each, yet contribute +0.008 to +0.060. These are the sources that cover the tail of the label space; the broad-coverage locations that score well alone sample the same head classes as each other, so their marginal value is negative once one of them is already active. Location 4 is the clear case: the highest solo profit in the pool (0.1095), 21 classes, and a marginal contribution of −0.051.

D.3 Amazon: label mismatch

Splice keeps 3 of 24 categories. The two highest-solo categories are both discarded (Industrial & Scientific at 0.829, Health & Household at 0.817), while the selected three rank 3rd, 4th and 14th. Solo profit spans a narrow 0.713–0.829 across the pool, so it separates categories weakly; marginal contributions span −0.030 to +0.036 and do not track it. Label mismatch is likewise not a sufficient

Table 13: Amazon source profiles, ordered by solo profit. *Pos.*: positive-class rate.

Category	Sel.	Pos.	Mismatch	Solo	Marginal
Industrial & Sci.		0.842	0.027	0.829	−0.008
Health & Household		0.812	0.003	0.817	−0.028
Kindle Store	✓	0.798	0.017	0.807	+0.036
Grocery & Gourmet	✓	0.830	0.015	0.805	+0.012
Software		0.688	0.127	0.804	−0.030
Home & Kitchen		0.828	0.013	0.791	−0.016
Patio, Lawn & Garden		0.806	0.009	0.789	−0.019
Tools & Home Impr.		0.830	0.015	0.788	−0.010
Books		0.828	0.013	0.787	−0.001
Office Products		0.870	0.055	0.784	−0.004
Arts, Crafts & Sewing		0.866	0.051	0.784	−0.011
Electronics		0.848	0.033	0.779	−0.025
Video Games		0.798	0.017	0.773	−0.023
Musical Instruments	✓	0.824	0.009	0.767	+0.010
CDs & Vinyl		0.912	0.097	0.763	−0.012
Sports & Outdoors		0.848	0.033	0.762	−0.012
Baby Products		0.832	0.017	0.759	−0.010
Automotive		0.840	0.025	0.757	−0.004
Clothing, Shoes & Jew.		0.856	0.041	0.757	−0.014
Appliances		0.846	0.031	0.746	−0.003
Pet Supplies		0.790	0.025	0.735	−0.012
Subscription Boxes		0.654	0.161	0.733	−0.013
Beauty & Personal Care		0.786	0.029	0.726	−0.019
Toys & Games		0.872	0.057	0.713	−0.015
Mean, selected (3)		0.817	0.014	0.793	+0.020
Mean, discarded (21)		0.826	0.042	0.770	−0.014

criterion on its own: Software and Subscription Boxes have the two largest mismatches (0.127, 0.161) and are discarded, but so are eighteen categories with mismatch below 0.06.